

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Mathematics — SSC-I (9th Grade)

Syllabus: Chapter 4 — Factorization and Algebraic Manipulation

Total Marks: 65

ANSWER KEY (Version 1)

Time: 2h 30m

SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

#	Ans	Correct Option	Explanation
1	B	$k(a + b + c)$	The common factor k is taken out from each term: $ka + kb + kc = k(a + b + c)$. This is the basic common factor type of factorization (Type recalled from previous grades).
2	C	$(a + b)(a - b)$	This is the standard difference of two squares identity: $a^2 - b^2 = (a + b)(a - b)$. Both binomials are factors; squaring option A gives $a^2 + 2ab + b^2$, which is wrong.
3	B	$(a^2 + ab + b^2)$ $(a^2 - ab + b^2)$	By adding and subtracting a^2b^2 : $a^4 + a^2b^2 + b^4 = (a^2 + b^2)^2 - (ab)^2$, which factors as $(a^2 + ab + b^2)(a^2 - ab + b^2)$. This is the Type-I factorization form from Section 4.1.
4	A	$(x - 1)(x - 8)$	Splitting the middle term: $x^2 - 9x + 8 = x^2 - x - 8x + 8 = x(x-1) - 8(x-1) = (x-1)(x-8)$. Product = 8, sum = 9, both negative, so factors -1 and -8 .
5	C	$(a + b)(a^2 - ab + b^2)$	The sum of two cubes identity is: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$. Note the middle sign inside the trinomial is negative ($-ab$), which distinguishes it from Option A.
6	B	$(a + b)^3$	The perfect cube expansion: $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$. This is the Type-IV form studied in Section 4.1 using the binomial cube identity.
7	C	Highest power of every factor	Per the textbook Key Fact (Section 4.3): HCF uses the lowest power of common factors, while LCM uses the highest power of every factor (both common and uncommon) that appears in any polynomial.
8	B	$P \times Q \div H$	From the relation proved in Section 4.3.2: $LCM \times HCF = P \times Q$, so $LCM = (P \times Q) \div HCF$. This is the fundamental HCF-LCM relationship used throughout the chapter.
9	C	$Q \neq 0$	An algebraic fraction P/Q is defined only when the denominator Q is non-zero ($Q \neq 0$), just as in arithmetic fractions. Division by zero is undefined in mathematics.
10	A	$\pm(2a + 3b)$	$4a^2 + 12ab + 9b^2 = (2a)^2 + 2(2a)(3b) + (3b)^2 = (2a + 3b)^2$. Applying square root gives $\pm(2a + 3b)$. The Key Fact (Section 4.5) states square roots always come in additive inverse pairs .
11	B	LCM of denominators	Section 4.4.4 states: algebraic fractions with different denominators are added by finding the LCM of the denominators (the common denominator), then combining numerators. HCF of denominators would give too small a value.
12	C	Multiplication	Section 4.1 explicitly states: "the factorization process is essentially the opposite of the multiplication process ." Multiplying factors gives the polynomial; factorizing the polynomial recovers the factors.

SECTION B — Short Question Answers (11 × 3 = 33 Marks)

Q. 2(i) 3 marksFactorize: $2x^2y^3 - 6x^2y^2 + 2xy^3$

$$\begin{aligned}
 &2x^2y^3 - 6x^2y^2 + 2xy^3 \\
 &= 2xy^2(xy - 3x + y) && \text{[take out HCF = } 2xy^2\text{]} && [1 \text{ mark}] \\
 &= 2xy^2(xy + y - 3x) && \text{[rearrange]} \\
 &= 2xy^2[y(x + 1) - 3x] && \text{[group: } y \text{ is common in first two]} && [1 \text{ mark}] \\
 \\
 &\therefore 2x^2y^3 - 6x^2y^2 + 2xy^3 = 2xy^2(xy + y - 3x) && [1 \text{ mark}]
 \end{aligned}$$

ORFactorize by grouping: $3nx - 3x - 3ny + 3y$

$$\begin{aligned}
 &3nx - 3x - 3ny + 3y \\
 &= 3(nx - x - ny + y) && \text{[take out common factor 3]} && [1 \text{ mark}] \\
 &= 3[x(n - 1) - y(n - 1)] && \text{[group: (n-1) is common]} && [1 \text{ mark}] \\
 &= 3(n - 1)(x - y) && [1 \text{ mark}]
 \end{aligned}$$

Q. 2(ii) 3 marksFactorize: $x^2 - 6ax + 9a^2 - 16b^2$

$$\begin{aligned}
 &x^2 - 6ax + 9a^2 - 16b^2 \\
 &= (x^2 - 6ax + 9a^2) - 16b^2 && \text{[group first three terms]} && [1 \text{ mark}] \\
 &= (x - 3a)^2 - (4b)^2 && \text{[perfect square trinomial]} && [1 \text{ mark}] \\
 &= (x - 3a + 4b)(x - 3a - 4b) && \text{[difference of squares]} && [1 \text{ mark}]
 \end{aligned}$$

ORFactorize: $1 - x^2 - 2xy - y^2$

$$\begin{aligned}
 &1 - x^2 - 2xy - y^2 \\
 &= 1 - (x^2 + 2xy + y^2) && \text{[group last three terms]} && [1 \text{ mark}] \\
 &= 1 - (x + y)^2 && \text{[perfect square]} && [1 \text{ mark}] \\
 &= [1 + (x + y)][1 - (x + y)] \\
 &= (1 + x + y)(1 - x - y) && \text{[difference of squares]} && [1 \text{ mark}]
 \end{aligned}$$

Q. 2(iii) **3 marks***Factorize: $x^4 + 4m^4$ (Type-I form)*

$$x^4 + 4m^4 = (x^2)^2 + (2m^2)^2$$

Add and subtract $4x^2m^2$ (the zero term): [1 mark]

$$= (x^2)^2 + 4x^2m^2 + (2m^2)^2 - 4x^2m^2$$

$$= (x^2 + 2m^2)^2 - (2xm)^2 \quad \text{[perfect square minus square]}$$

$$= (x^2 + 2m^2 + 2xm)(x^2 + 2m^2 - 2xm) \quad \text{[1 mark]}$$

$$\therefore x^4 + 4m^4 = (x^2 + 2mx + 2m^2)(x^2 - 2mx + 2m^2) \quad \text{[1 mark]}$$

OR*Factorize: $m^4 + m^2 + 1$*

$$m^4 + m^2 + 1 = (m^2)^2 + m^2 + 1$$

Add and subtract m^2 : [1 mark]

$$= (m^2)^2 + 2m^2 + 1 - m^2$$

$$= (m^2 + 1)^2 - (m)^2 \quad \text{[difference of squares]} \quad \text{[1 mark]}$$

$$= (m^2 + 1 + m)(m^2 + 1 - m)$$

$$= (m^2 + m + 1)(m^2 - m + 1) \quad \text{[1 mark]}$$

Q. 2(iv) **3 marks***Factorize: $y^2 - 7y + 12$*

$$y^2 - 7y + 12$$

Factors of 12 whose sum = -7: -3 and -4 [1 mark]

$$= y^2 - 4y - 3y + 12$$

$$= y(y - 4) - 3(y - 4) \quad \text{[group]} \quad \text{[1 mark]}$$

$$= (y - 4)(y - 3) \quad \text{[1 mark]}$$

OR*Factorize: $3x^2 + 22x - 16$*

$$3x^2 + 22x - 16$$

Product of extreme coefficients: $3 \times (-16) = -48$

Factors of -48 whose sum = +22: +24 and -2 [1 mark]

$$= 3x^2 + 24x - 2x - 16$$

$$= 3x(x + 8) - 2(x + 8) \quad \text{[group]} \quad \text{[1 mark]}$$

$$= (x + 8)(3x - 2) \quad \text{[1 mark]}$$

Q. 2(v) 3 marksFactorize: $x^3 + 27y^3$

$$x^3 + 27y^3 = (x)^3 + (3y)^3 \quad \text{[identify cubes]} \quad [1 \text{ mark}]$$

$$\text{Using: } a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$\text{with } a = x, b = 3y: \quad [1 \text{ mark}]$$

$$= (x + 3y)[(x)^2 - (x)(3y) + (3y)^2]$$

$$= (x + 3y)(x^2 - 3xy + 9y^2) \quad [1 \text{ mark}]$$

ORFactorize: $8a^3 - 125b^3 - 2a + 5b$

$$8a^3 - 125b^3 - 2a + 5b$$

$$= (2a)^3 - (5b)^3 - (2a - 5b) \quad \text{[rewrite]} \quad [1 \text{ mark}]$$

$$= (2a - 5b)[(2a)^2 + (2a)(5b) + (5b)^2] - (2a - 5b)$$

$$[a^3 - b^3 = (a - b)(a^2 + ab + b^2)] \quad [1 \text{ mark}]$$

$$= (2a - 5b)[4a^2 + 10ab + 25b^2 - 1] \quad \text{[factor out (2a-5b)]} \quad [1 \text{ mark}]$$

Q. 2(vi) 3 marksFactorize: $27a^3 + 1 + 27a^2 + 9a$

$$27a^3 + 27a^2 + 9a + 1 \quad \text{[rearrange]}$$

$$= (3a)^3 + 3(3a)^2(1) + 3(3a)(1)^2 + (1)^3 \quad \text{[identify binomial cube]} \quad [1 \text{ mark}]$$

$$= (3a + 1)^3 \quad \text{[using (a+b)^3 expansion]} \quad [1 \text{ mark}]$$

$$\therefore 27a^3 + 1 + 27a^2 + 9a = (3a + 1)^3 \quad [1 \text{ mark}]$$

ORFactorize: $4x^2(2x - 15) - 25(5 - 6x)$

$$4x^2(2x - 15) - 25(5 - 6x)$$

$$= 8x^3 - 60x^2 - 125 + 150x \quad \text{[expand]} \quad [1 \text{ mark}]$$

$$= 8x^3 - 60x^2 + 150x - 125 \quad \text{[rearrange]}$$

$$= (2x)^3 - 3(2x)^2(5) + 3(2x)(5)^2 - (5)^3 \quad [1 \text{ mark}]$$

$$= (2x - 5)^3 \quad \text{[(a-b)^3 form]} \quad [1 \text{ mark}]$$

Q. 2(vii) 3 marks

Find the HCF of: $(x + y)^2$ and $x^2 - y^2$

Factorize each expression:

$$(x + y)^2 = (x + y)(x + y) \quad \text{[complete factored form]} \quad [1 \text{ mark}]$$

$$x^2 - y^2 = (x + y)(x - y) \quad \text{[difference of squares]} \quad [1 \text{ mark}]$$

Common factor with least power = $(x + y)^1$

$$\therefore \text{HCF} = (x + y) \quad [1 \text{ mark}]$$

OR

Find the HCF of: $(a - b)^3$ and $a^2 - 2ab + b^2$

Factorize each expression:

$$(a - b)^3 = (a - b)^3 \quad \text{[already factored]} \quad [1 \text{ mark}]$$

$$a^2 - 2ab + b^2 = (a - b)^2 \quad \text{[perfect square]} \quad [1 \text{ mark}]$$

Lowest power of common factor $(a - b)$:

$$\min(3, 2) = 2$$

$$\therefore \text{HCF} = (a - b)^2 \quad [1 \text{ mark}]$$

Q. 2(viii) 3 marks

Find the HCF of: $ax^2 + 7ax + 12a$, $ax^2 - 5ax - 24a$ and $2ax^2 + 5ax - 3a$

Factor each:

$$ax^2 + 7ax + 12a = a(x^2 + 7x + 12) = a(x+3)(x+4) \quad [1 \text{ mark}]$$

$$ax^2 - 5ax - 24a = a(x^2 - 5x - 24) = a(x-8)(x+3) \quad [1 \text{ mark}]$$

$$2ax^2 + 5ax - 3a = a(2x^2 + 5x - 3) = a(2x-1)(x+3)$$

Common factors with least power: a^1 and $(x+3)^1$

$$\therefore \text{HCF} = a(x + 3) \quad [1 \text{ mark}]$$

OR

Find the HCF of $x^2 - 49$ and $x^2 - 4x - 21$

$$x^2 - 49 = (x + 7)(x - 7) \quad \text{[difference of squares]} \quad [1 \text{ mark}]$$

$$\begin{aligned} x^2 - 4x - 21: & \text{ factors of } -21 \text{ with sum } -4 \rightarrow -7 \text{ and } +3 \\ & = (x - 7)(x + 3) \end{aligned} \quad [1 \text{ mark}]$$

Common factor = $(x - 7)$

$$\therefore \text{HCF} = (x - 7) \quad [1 \text{ mark}]$$

Q. 2(ix) **3 marks**

State $LCM \times HCF = P \times Q$. If $HCF = x - 7$ and $LCM = x^3 - 10x^2 + 11x + 70$, find the product of the two polynomials.

Key relation: $LCM \times HCF = P \times Q$ [1 mark]

$$HCF = (x - 7)$$

$$LCM = x^3 - 10x^2 + 11x + 70$$

Product $P \times Q = LCM \times HCF$ [1 mark]

$$= (x^3 - 10x^2 + 11x + 70)(x - 7)$$

(Simplification not required – the relation gives the product directly)

$$\therefore P \times Q = (x - 7)(x^3 - 10x^2 + 11x + 70)$$
 [1 mark]

OR

Find the LCM of: $x^2 + x - 6$, $x^2 + x - 2$ and $x^2 - 4x + 3$

Factorize each:

$$x^2 + x - 6 = (x + 3)(x - 2)$$
 [1 mark]

$$x^2 + x - 2 = (x + 2)(x - 1)$$

Wait – let me verify: $x^2 + x - 6$: need sum +1, product -6 → +3 and -2 ✓

$x^2 + x - 2$: sum +1, product -2 → +2 and -1 ✓

$x^2 - 4x + 3$: sum -4, product +3 → -3 and -1 → $(x-3)(x-1)$ [1 mark]

All factors: $(x+3)$, $(x-2)$, $(x+2)$, $(x-1)$, $(x-3)$

LCM = product of all distinct factors (each to highest power)

$$\therefore LCM = (x + 3)(x + 2)(x - 1)(x - 2)(x - 3)$$
 [1 mark]

Q. 2(x) 3 marks

Simplify: $(x^2 - 1)/(x^2 + 4x + 4) \times (x + 2)/(x^2 + 2x - 3)$

Step 1: Factorize all numerators and denominators:

$$x^2 - 1 = (x+1)(x-1)$$

$$x^2 + 4x + 4 = (x+2)(x+2)$$

$$x + 2 = (x+2)$$

$$x^2 + 2x - 3 = (x+3)(x-1) \quad [1 \text{ mark}]$$

Step 2: Write as one fraction and cancel:

$$= [(x+1)(x-1)] / [(x+2)^2] \times (x+2) / [(x+3)(x-1)]$$

$$= [(x+1)(x-1)(x+2)] / [(x+2)^2(x+3)(x-1)] \quad [1 \text{ mark}]$$

$$= (x+1) / [(x+2)(x+3)]$$

$$\therefore \text{Result} = (x + 1) / (x^2 + 5x + 6) \quad [1 \text{ mark}]$$

OR

Simplify: $(r^2 - s^2)/r \div (r - s)/s$

$$= (r^2 - s^2)/r \times s/(r - s) \quad [\text{invert divisor}] \quad [1 \text{ mark}]$$

$$= [(r+s)(r-s)] / r \times s/(r-s) \quad [\text{factorize } r^2-s^2] \quad [1 \text{ mark}]$$

$$= s(r+s) / r \quad [\text{cancel } (r-s)]$$

$$\therefore \text{Result} = (sr + s^2) / r \quad [1 \text{ mark}]$$

Q. 2(xi) 3 marks

Simplify: $1/[x(x - y)] + 1/[y(x + y)]$

$$\text{LCM of denominators} = xy(x - y)(x + y) \quad [1 \text{ mark}]$$

$$= [y(x+y) + x(x-y)] / [xy(x-y)(x+y)] \quad [1 \text{ mark}]$$

$$= [xy + y^2 + x^2 - xy] / [xy(x^2 - y^2)]$$

$$= [x^2 + y^2] / [xy(x^2 - y^2)]$$

$$\therefore \text{Result} = (x^2 + y^2) / [xy(x^2 - y^2)] \quad [1 \text{ mark}]$$

OR

Find the square root of $4a^2 + 12ab + 9b^2$ by factorization method.

$$4a^2 + 12ab + 9b^2$$

$$= (2a)^2 + 2(2a)(3b) + (3b)^2 \quad [\text{identify perfect square form}] \quad [1 \text{ mark}]$$

$$= (2a + 3b)^2 \quad [\text{complete square}] \quad [1 \text{ mark}]$$

Applying square root:

$$\sqrt{(4a^2 + 12ab + 9b^2)} = \sqrt{(2a + 3b)^2} = \pm(2a + 3b) \quad [1 \text{ mark}]$$

SECTION C — Long Question Answers ($4 \times 5 = 20$ Marks)

Q. 3(i) 5 marks

Factorize completely: $(x^2 + 3x - 4)(x^2 + 3x + 5) + 8$

Observe that the first two terms inside both brackets are identical: $x^2 + 3x$

Step 1: Let $a = x^2 + 3x$ [1 mark]

Then the expression becomes:

$$(a - 4)(a + 5) + 8$$

Step 2: Expand:

$$= a^2 + 5a - 4a - 20 + 8$$

$$= a^2 + a - 12 \quad [1 \text{ mark}]$$

Step 3: Factorize $a^2 + a - 12$:

Factors of -12 with sum $+1$: $+4$ and -3

$$= (a + 4)(a - 3) \quad [1 \text{ mark}]$$

Step 4: Substitute back $a = x^2 + 3x$:

$$= (x^2 + 3x + 4)(x^2 + 3x - 3) \quad [1 \text{ mark}]$$

Verify neither factor factorizes further (discriminants < 0 for first):

$$\text{Discriminant of } x^2 + 3x + 4: 9 - 16 = -7 < 0 \text{ (irreducible) } \checkmark$$

$$\therefore (x^2 + 3x - 4)(x^2 + 3x + 5) + 8 = (x^2 + 3x + 4)(x^2 + 3x - 3) \quad [1 \text{ mark}]$$

OR

Factorize completely: $(x + 5)(x + 3)(x + 2)(x + 6) - 88$

Step 1: Pair the binomials strategically.

Notice: $5 + 3 = 8$ and $2 + 6 = 8$ (sums are equal) [1 mark]

So pair $(x+5)(x+3)$ and $(x+2)(x+6)$:

$$(x+5)(x+3) = x^2 + 8x + 15$$

$$(x+2)(x+6) = x^2 + 8x + 12$$

Step 2: Let $a = x^2 + 8x$ [1 mark]

$$= (a + 15)(a + 12) - 88$$

Step 3: Expand:

$$= a^2 + 12a + 15a + 180 - 88$$

$$= a^2 + 27a + 92 \quad [1 \text{ mark}]$$

Step 4: Factorize $a^2 + 27a + 92$:

Factors of 92 with sum 27 : $+4$ and $+23$

$$= (a + 4)(a + 23) \quad [1 \text{ mark}]$$

Step 5: Substitute back $a = x^2 + 8x$:

$$= (x^2 + 8x + 4)(x^2 + 8x + 23) \quad [1 \text{ mark}]$$

$$\therefore (x+5)(x+3)(x+2)(x+6) - 88 = (x^2 + 8x + 4)(x^2 + 8x + 23)$$

Q. 3(ii) 5 marks

Find the HCF of $P = 2m^2 - 2mn$, $Q = 4m^4 - 4m^2n^2$, $R = 2m^3 - 4m^2n + 2mn^2$ by factorization.

Step 1: Factorize P:

$$2m^2 - 2mn = 2m(m - n) \quad [1 \text{ mark}]$$

Step 2: Factorize Q:

$$\begin{aligned} 4m^4 - 4m^2n^2 &= 4m^2(m^2 - n^2) = 4m^2(m+n)(m-n) \\ &= 2^2 \cdot m^2 \cdot (m+n) \cdot (m-n) \end{aligned} \quad [1 \text{ mark}]$$

Step 3: Factorize R:

$$\begin{aligned} 2m^3 - 4m^2n + 2mn^2 \\ &= 2m(m^2 - 2mn + n^2) \\ &= 2m(m - n)^2 \end{aligned} \quad [1 \text{ mark}]$$

Step 4: List common factors with LOWEST powers:

$$P = 2 \cdot m \cdot (m-n)$$

$$Q = 2^2 \cdot m^2 \cdot (m+n) \cdot (m-n)$$

$$R = 2 \cdot m \cdot (m-n)^2$$

Common numerical: 2^1 (lowest power of 2)

Common variable: m^1 (lowest power of m)

Common binomial: $(m-n)^1$ (lowest power of m-n) [1 mark]

$$\therefore \text{HCF} = 2m(m - n) \quad [1 \text{ mark}]$$

OR

Find the HCF of $P = x^3 + x^2 - 5x + 3$ and $Q = x^2 + 3x$ by the division method.

Step 1: P has higher degree; use $Q = x^2 + 3x$ as divisor.

Note: $x^2 + 3x = x(x + 3)$, degree 2; P degree 3. [1 mark]

Divide P by Q:

$$x^3 + x^2 - 5x + 3 \div (x^2 + 3x)$$

$$\begin{array}{r} x^3 + x^2 - 5x + 3 \\ -(x^3 + 3x^2) \\ \hline \end{array}$$

$$\begin{array}{r} -2x^2 - 5x + 3 \\ -(-2x^2 - 6x) \\ \hline \end{array}$$

$$x + 3 \quad \leftarrow \text{remainder } R_1 \quad [1 \text{ mark}]$$

Step 2: Now divide Q by R_1 :

$$x^2 + 3x \div (x + 3)$$

$$= x(x + 3) \div (x + 3)$$

$$= x \quad \text{remainder } 0$$

[1 mark]

[1 mark]

Since remainder is 0, last divisor = HCF.

$$\therefore \text{HCF} = x + 3 \quad [1 \text{ mark}]$$

Q. 3(iii) 5 marks

Find the LCM of $2x^2 - 12xy + 16y^2$, $x^2 - 6xy + 8y^2$ and $3x^2 - 12y^2$. Hence verify $\text{LCM} \times \text{HCF} = P \times Q$

for the first two.

Step 1: Factorize all three:

$$2x^2 - 12xy + 16y^2 = 2(x^2 - 6xy + 8y^2) = 2(x-4y)(x-2y) \quad [1 \text{ mark}]$$

$$x^2 - 6xy + 8y^2 = (x-4y)(x-2y)$$

$$3x^2 - 12y^2 = 3(x^2 - 4y^2) = 3(x+2y)(x-2y) \quad [1 \text{ mark}]$$

Step 2: All distinct factors (with highest powers):

$$2, 3, (x-4y), (x-2y), (x+2y)$$

$$\begin{aligned} \text{LCM} &= 2 \times 3 \times (x-4y) \times (x-2y) \times (x+2y) \\ &= 6(x-4y)(x-2y)(x+2y) \end{aligned} \quad [1 \text{ mark}]$$

Step 3: Find HCF of first two (for verification):

$$P = 2(x-4y)(x-2y)$$

$$Q = (x-4y)(x-2y)$$

$$\text{Common factors (lowest power): } (x-4y)^1(x-2y)^1$$

$$\text{HCF} = (x-4y)(x-2y) \quad [1 \text{ mark}]$$

Step 4: Verify $\text{LCM} \times \text{HCF} = P \times Q$:

$$\begin{aligned} \text{LCM} \times \text{HCF} &= 6(x-4y)(x-2y)(x+2y) \times (x-4y)(x-2y) \\ &= 6(x-4y)^2(x-2y)^2(x+2y) \end{aligned}$$

$$\begin{aligned} P \times Q &= 2(x-4y)(x-2y) \times (x-4y)(x-2y) \\ &= 2(x-4y)^2(x-2y)^2 \end{aligned}$$

These are NOT equal – the LCM in the verification should use only P and Q:

$$\text{LCM of P and Q} = 2(x-4y)(x-2y) = P \quad (\text{since P is a multiple of Q})$$

$$\begin{aligned} \text{LCM}(P, Q) \times \text{HCF}(P, Q) &= 2(x-4y)(x-2y) \times (x-4y)(x-2y) \\ &= 2(x-4y)^2(x-2y)^2 \end{aligned}$$

$$\begin{aligned} P \times Q &= 2(x-4y)(x-2y) \times (x-4y)(x-2y) \\ &= 2(x-4y)^2(x-2y)^2 \quad \checkmark \quad (\text{Both sides equal}) \end{aligned} \quad [1 \text{ mark}]$$

OR

Product of two polynomials is $x^4 + 3x^3 - 12x^2 - 20x + 48$ and their HCF is $x - 2$. Find their LCM. Also verify $\text{LCM} \times \text{HCF} = P \times Q$.

Given: $P \times Q = x^4 + 3x^3 - 12x^2 - 20x + 48$
 $HCF = x - 2$

Step 1: Use $LCM = (P \times Q) / HCF$ [1 mark]

Step 2: Divide the product by $(x - 2)$:

$$x^4 + 3x^3 - 12x^2 - 20x + 48 \div (x - 2)$$

By synthetic division or long division:

$$x^4 + 3x^3 - 12x^2 - 20x + 48$$

$$-(x^4 - 2x^3)$$

$$5x^3 - 12x^2 - 20x + 48$$

$$-(5x^3 - 10x^2)$$

$$-2x^2 - 20x + 48$$

$$-(-2x^2 + 4x)$$

$$-24x + 48$$

$$-(-24x + 48)$$

$$0$$

[2 marks]

$$LCM = x^3 + 5x^2 - 2x - 24$$

[1 mark]

Step 3: Verify $LCM \times HCF = P \times Q$:

$$LCM \times HCF = (x^3 + 5x^2 - 2x - 24)(x - 2)$$

$$= x^4 + 3x^3 - 12x^2 - 20x + 48 = P \times Q \quad \checkmark$$

[1 mark]

Q. 3(iv) 5 marks

Simplify: $-2x/(x+3) + 3/(3-x) - (8x-12)/(x^2-9)$

Step 1: Factorize denominators:

$$x + 3 = (x + 3)$$

$$3 - x = -(x - 3)$$

$$x^2 - 9 = (x + 3)(x - 3) \quad [1 \text{ mark}]$$

Step 2: Rewrite with consistent signs:

$$= -2x/(x+3) + 3/(-(x-3)) - (8x-12)/[(x+3)(x-3)]$$

$$= -2x/(x+3) - 3/(x-3) - (8x-12)/[(x+3)(x-3)] \quad [1 \text{ mark}]$$

Step 3: LCM of denominators = $(x+3)(x-3)$

$$= [-2x(x-3) - 3(x+3) - (8x-12)] / [(x+3)(x-3)] \quad [1 \text{ mark}]$$

Step 4: Expand numerator:

$$-2x(x-3) = -2x^2 + 6x$$

$$-3(x+3) = -3x - 9$$

$$-(8x-12) = -8x + 12$$

$$\begin{aligned} \text{Numerator} &= -2x^2 + 6x - 3x - 9 - 8x + 12 \\ &= -2x^2 - 5x + 3 \end{aligned} \quad [1 \text{ mark}]$$

Step 5: Factorize numerator:

$$-2x^2 - 5x + 3 = -1(2x^2 + 5x - 3) = -1(2x-1)(x+3)$$

$$= -(2x-1)(x+3) / [(x+3)(x-3)]$$

$$= -(2x-1) / (x-3)$$

$$= (1-2x)/(x-3) \text{ or } -(2x-1)/(x-3) \quad [1 \text{ mark}]$$

$$\therefore \text{Result} = -(2x - 1) / (x - 3)$$

OR

Simplify: $(3u^2+5uv+2v^2)/(u^2+5uv+6v^2) + (2u+6v)/(u^2-4v^2) \div (u^2+6uv+9v^2)/(u^2v-2uv^2)$

Step 1: Apply DMAS – handle division first:

$$\begin{aligned} & (2u+6v)/(u^2-4v^2) \div (u^2+6uv+9v^2)/(u^2v-2uv^2) \\ & = (2u+6v)/(u^2-4v^2) \times (u^2v-2uv^2)/(u^2+6uv+9v^2) \end{aligned} \quad [1 \text{ mark}]$$

Step 2: Factorize all:

$$2u+6v = 2(u+3v)$$

$$u^2-4v^2 = (u+2v)(u-2v)$$

$$u^2v-2uv^2 = uv(u-2v)$$

$$u^2+6uv+9v^2 = (u+3v)^2$$

$$\begin{aligned} & = 2(u+3v)/[(u+2v)(u-2v)] \times uv(u-2v)/(u+3v)^2 \\ & = 2uv(u+3v)(u-2v) / [(u+2v)(u-2v)(u+3v)^2] \\ & = 2uv / [(u+2v)(u+3v)] \end{aligned} \quad [1 \text{ mark}]$$

Step 3: Now add the first fraction:

$$(3u^2+5uv+2v^2)/(u^2+5uv+6v^2)$$

Factorize:

$$\begin{aligned} 3u^2+5uv+2v^2 &= (3u+2v)(u+v) \quad [\text{product 6, sum 5: } +3v \text{ and } +2v \dots \text{ wait}] \\ &= (u+v)(3u+2v) \quad \checkmark \end{aligned}$$

$$u^2+5uv+6v^2 = (u+2v)(u+3v) \quad [1 \text{ mark}]$$

$$= (u+v)(3u+2v)/[(u+2v)(u+3v)] + 2uv/[(u+2v)(u+3v)]$$

Step 4: Common denominator = $(u+2v)(u+3v)$:

$$= [(u+v)(3u+2v) + 2uv] / [(u+2v)(u+3v)]$$

Expand numerator:

$$\begin{aligned} (u+v)(3u+2v) &= 3u^2 + 2uv + 3uv + 2v^2 = 3u^2 + 5uv + 2v^2 \\ + 2uv &\rightarrow 3u^2 + 7uv + 2v^2 \end{aligned} \quad [1 \text{ mark}]$$

Step 5: Factorize $3u^2 + 7uv + 2v^2$:

$$\text{Product 6, sum 7: } +6 \text{ and } +1 \rightarrow (3u+v)(u+2v)$$

$$= (3u+v)(u+2v) / [(u+2v)(u+3v)]$$

$$= (3u+v) / (u+3v)$$

$$\therefore \text{Result} = (3u + v) / (u + 3v) \quad [1 \text{ mark}]$$